Poster Session - ACM IUI 2019

Relaxation “sweet spot” exploration in pantophonic musical soundscape using reinforcement learning

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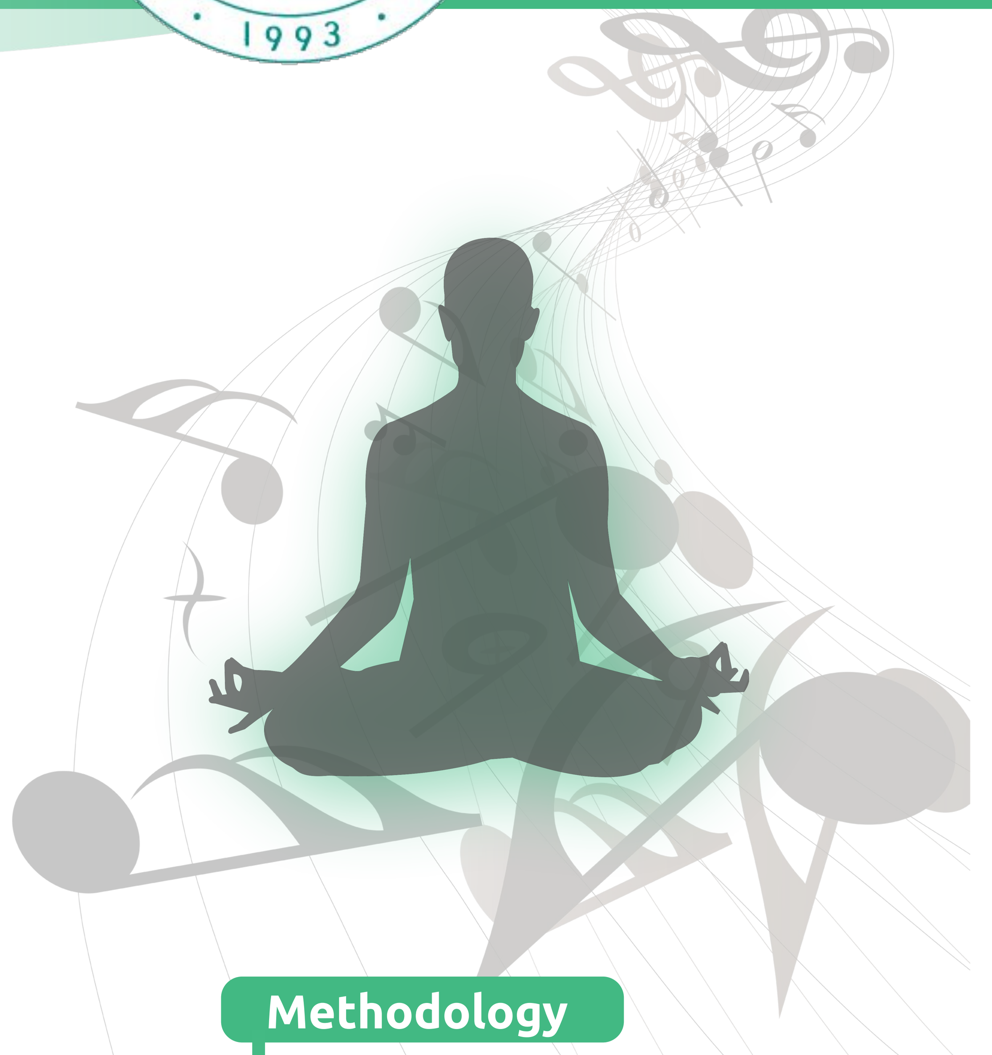

Introduction

- Relaxation techniques for reducing effects of stress can improve physical and mental health.
- Among many relaxation techniques, musical relaxation is one of the simplest.
- Nature sounds, instrumental music, voice (chanting), “easy listening” songs, etc. can be played for relaxation.
- Selecting suitable sounds for a particular subject is challenging because of idiosyncratic tastes and circumstances.
- In our approach, computer-guided audition for spatial soundscapes is investigated, automatically exploring a polyphonic area using biosignals as indicators of satisfaction.
- Reinforcement learning (RL) is a way of programming agents by rewards and penalties to achieve a goal.
- Reward values are calculated using change of relative theta band (4–8 Hz) power of electroencephalography (EEG) signals fed to a Deep Q-Network (DQN) RL algorithm to adjust the parameters of spatial soundscape.

Methodology

- As shown in Figure 1, a Max/MSP patch was developed as a soundscape interface.
- Headphones present this soundscape featuring six pantophonic sources arranged in a ring around the perimeter of the virtual space, which is parameterized not only by Cartesian position but also by processing attributes such as volume, pitch, reverb, and various parameters (center and cutoff frequencies, and resonance) for low-pass, high-pass, bandpass, and notch filtering.

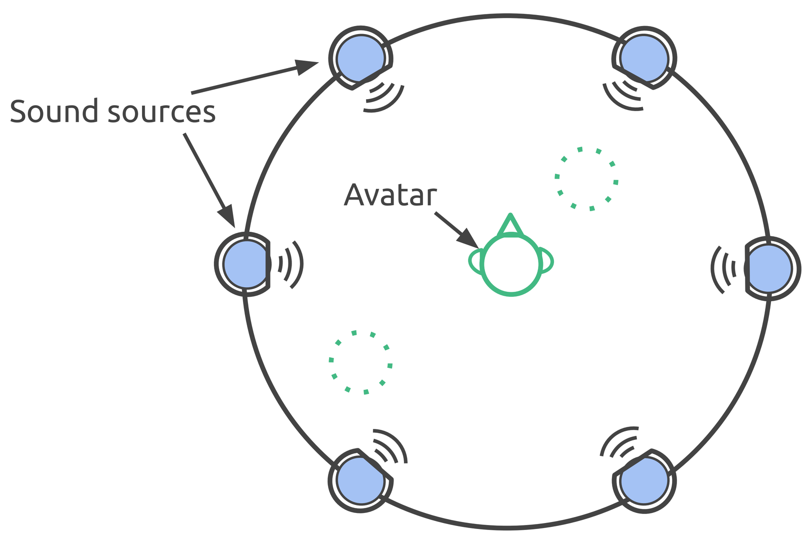

Fig. 1: Horizontal soundscape map showing iconic avatar and distribution of sound sources in Max/MSP interface

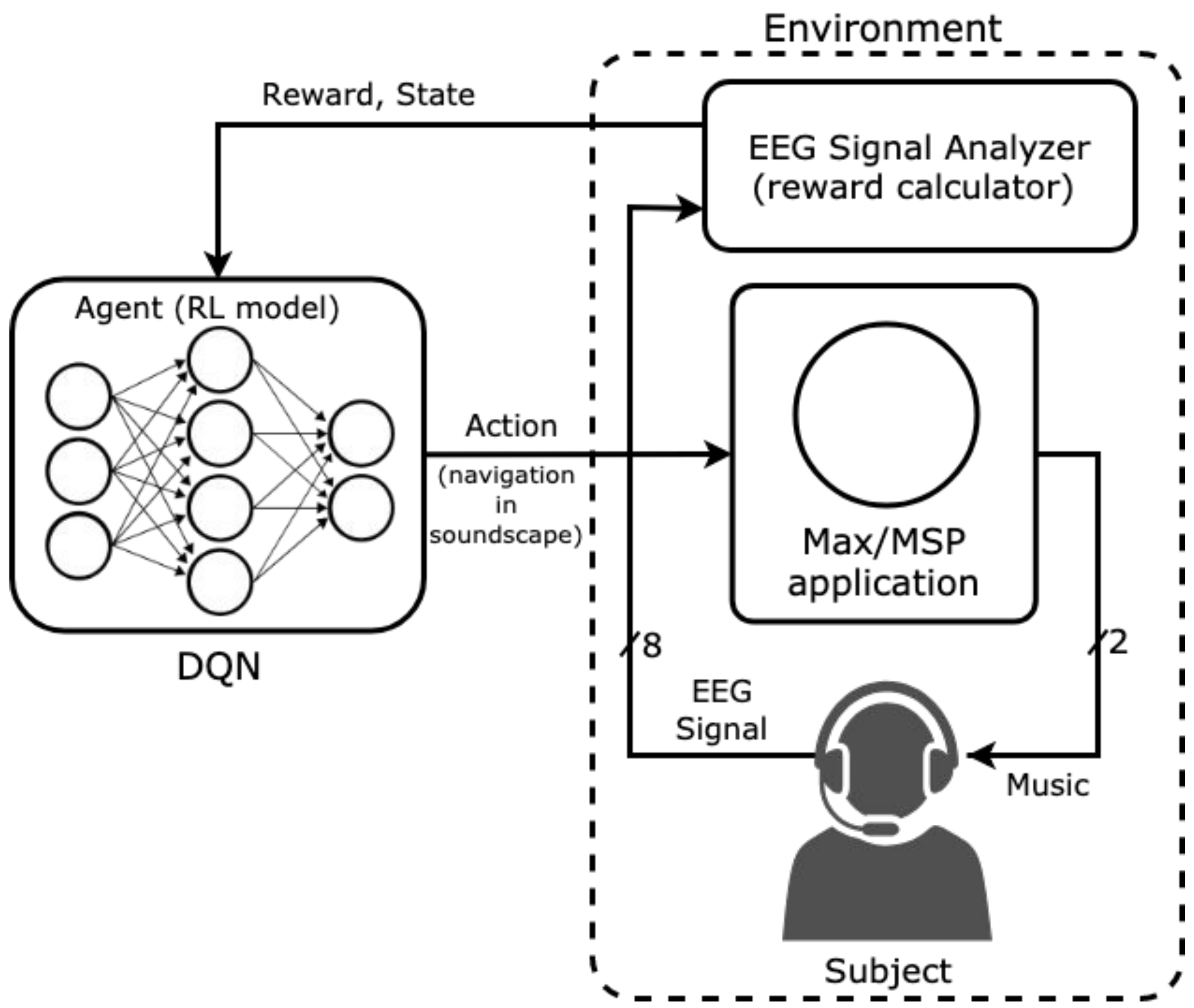

Fig. 2: System architecture

- “OpenBCI Cyton” 3D-printed helmet as shown in Figure 3, with 8 mounted EEG sensors, was used to collect data.
- At run-time, the subject was instructed to keep eyes closed to minimize distractions, and auditioned the multidimensional soundscape.
- When a person relaxes, the theta band (4–8 Hz) power of EEG signal increases accordingly.
- The realtime data stream is broken into windows with 50% overlap and filtered using a Butterworth band-pass filters to extract the relative theta frequency band power.
- By analysing the trend of band power, corresponding reward values were calculated.
- As shown in Figure 2, the DQN algorithm takes the reward and state parameters from the environment and changes the position of the avatar.

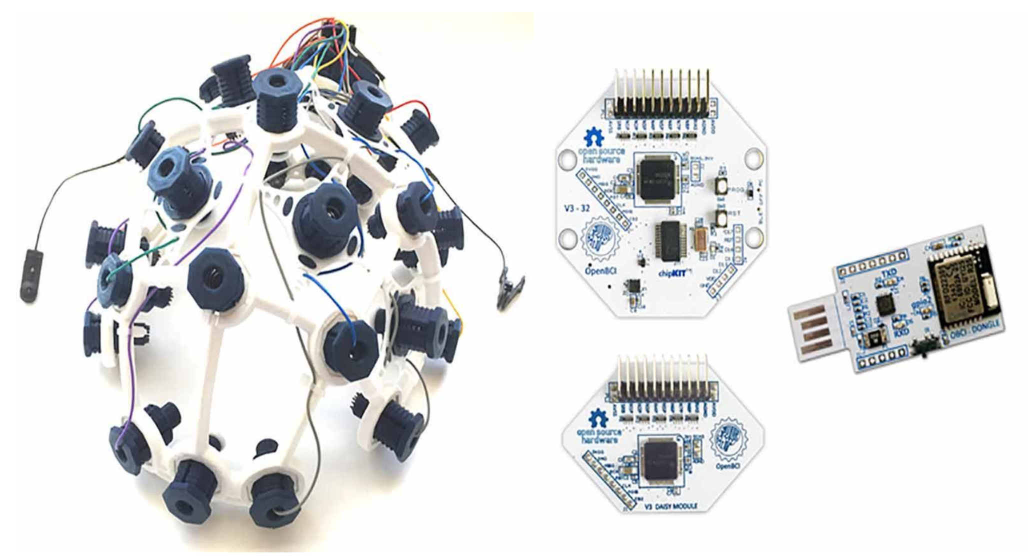

Fig. 3: OpenBCI Cyton headset

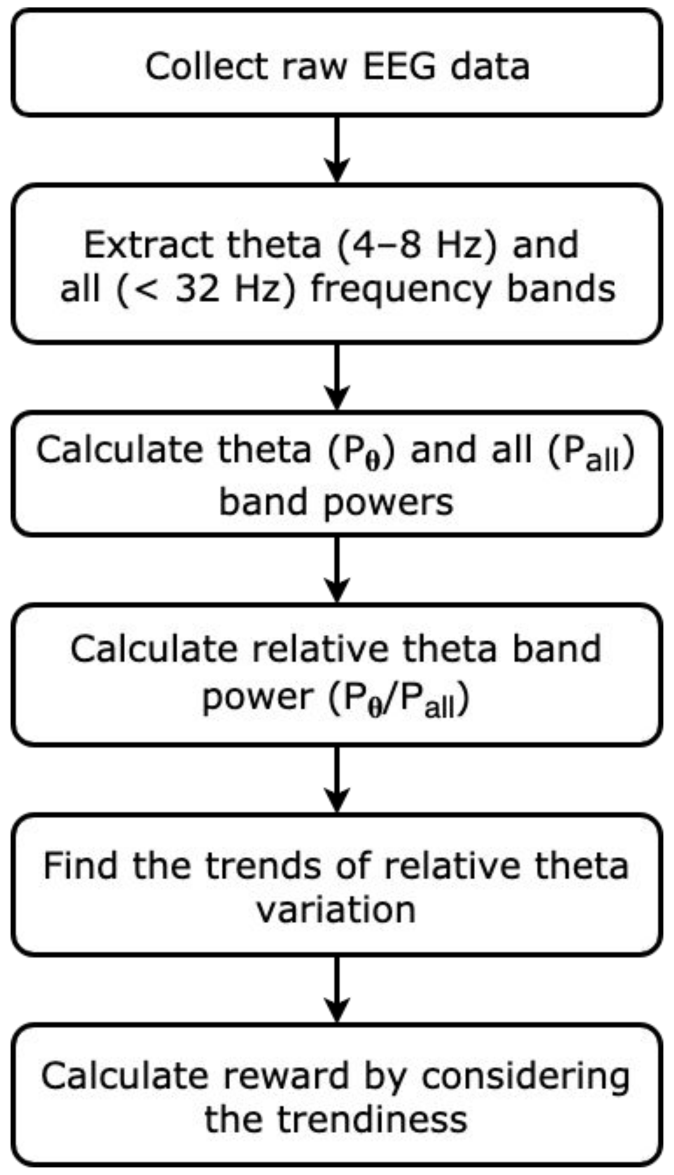

Fig. 4: Reward calculation

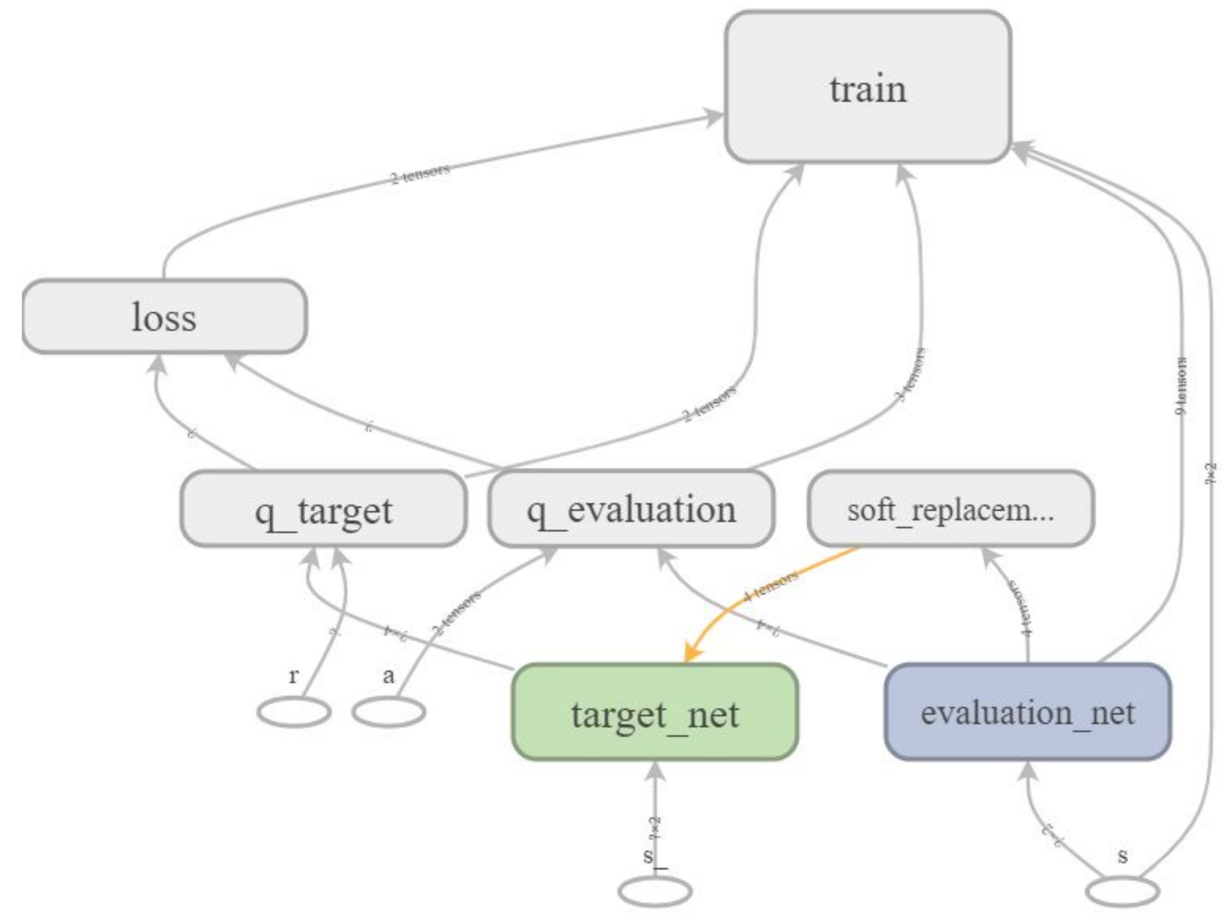

Fig. 5: TensorBoard-generated DQN structure

Discussion

- Our experiment assumes that relaxation can be described by a stable field across the pantophonic soundscape bounded by the annulus (ring) of a half-dozen musical sources.
- The sweet spot is probably not a point, but an area or collection of locations (perhaps unconnected).
- There are also several reasons the sweet spot probably isn’t stable, including memory & habituation, boredom, hysteresis, and fatigue.
- EEG signals might not be the best parameter for measuring relaxation, but we leverage our experience with such biosensing, rather than, for instance, self-reported satisfaction (which could also interfere with such psychological state).

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